

Hasselt University Belgium
Department of Biostatistics and Data Science
Masters Thesis Topic Proposal Abstract

Prepared by:

Langsi Ambe Revelation (final year student in Data Science)

Machine Learning in Football Prediction for the Belgian Jupiler Pro League

Football matches have many variables that affect the result of a game, such as team performance, player form, match circumstances, and uncontrollable incidents like injuries or referee decisions. For this reason, predicting the result of a game has always been difficult. Even with the abundance of historical data available, football is still a game where a lot of luck plays a part, which makes forecasting outcomes challenging. By creating a machine learning (ML) model specially made to forecast the outcomes of games in the Belgian Jupiler Pro League, focussing on three possible outcomes—a home team win, a draw, or an away team win—this research aims to address this difficulty.

To produce informative and precise pre-match predictions, the objective is to build a strong and trustworthy predictive model that can be used to analyze past match data, individual statistics, and team measures. This project seeks to increase predicted accuracy and add to the expanding field of sports analytics research through the application of a variety of machine learning methods. With sports being inherently unpredictable, the suggested machine learning model would make use of data analysis and machine learning techniques like random forests, logistic regression, and deep learning to increase the possibility of generating reliable predictions.

Methodology

The project will be carried out in multiple stages, using various techniques for data analysis and machine learning in each phase:

1. Data mining and preprocessing

- **Data Sources:** Historical match data, player statistics, team performance metrics, weather conditions, and other relevant factors will be collected from sources such as Kaggle, Football-Data.co.uk, and official league websites.
- **Data Cleaning:** Handling missing values, outliers, and ensuring data consistency.
- **Feature Engineering:** Creating new features such as form indicators, head-to-head statistics, and player fitness levels.

2. Exploratory Data Analysis

- **Descriptive Statistics:** summarising the main characteristics of the data.
- **Visualization:** Finding patterns and correlations with the use of graphs and plots.

3. Model Selection and Training

To determine the most accurate method for match outcome prediction, the project will investigate and contrast a number of machine learning models. Several of the main techniques consist of:

a. Supervised Learning Algorithms:

- **Multinomial Logistics Regression:** a simple categorization model that will serve as the project's starting point. It is capable of accurately estimating the odds of winning at home, drawing, or winning away.
-
- **Support Vector Machine (SVM):** It is appropriate for complicated datasets since it can use a non-linear decision boundary to divide the three possible outcomes (home victory, draw, and away win).
-
- **Random Forest:** An ensemble method that constructs multiple decision trees and aggregates their predictions to improve accuracy. It handles large datasets and prevents overfitting.
- **XGBoost:** A highly efficient gradient-boosting algorithm known for high accuracy. It is an ideal method for handling complicated data and unbalanced results since it enhances model performance iteratively.
- **Neural Network (Deep Learning):** Neural networks, especially long short-term memory (LSTM) networks and recurrent neural networks (RNNs), can be used to manage time-series data (e.g., recent performance trends) in order to gain deeper insights.

b. Unsupervised Learning Algorithms

- **K-Nearest Neighbours (KNN):** This technique looks for patterns in recent games and categorizes results based on how similar they were to matches in the past.
- **Naive Base:** Probabilistic models that are capable of accounting for uncertainty and predicting outcomes based on the relationships between factors such as team makeup and game circumstances.
- **Ensemble learning:** combining several models to enhance accuracy in predictions

4. Model Evaluation:

Each model's performance will be assessed using ROC-AUC, F1-score, accuracy, precision, confusion matrix (to see actual true and false predictions), and recall. Model generalization and the prevention of overfitting will be aided by cross-validation, especially k-fold validation.

5. Challenges and Limitations:

- Because football is an unpredictable sport, predictions about its results are always uncertain, even with breakthroughs in machine learning. Unpredictability is introduced by elements like weather, injuries, and referee rulings, which models might not entirely account for.

- The quantity and quality of the available data will have a significant impact on the accuracy of the model, particularly when it comes to individual or team statistics that are erroneous or incomplete.

6. Expected outcome:

It is anticipated that the research will produce a strong machine learning model that can predict football match results in the Belgian Jupiler Pro League with accuracy. The model's insights can be applied to betting, strategic planning, and raising fan involvement.

7. Data sources

- **Kaggle**
- **Football-Data.co.uk**
- **Official league websites and databases**

This thesis will contribute to the growing field of sports analytics by demonstrating the application of machine learning in football prediction, specifically within the context of the Belgian Jupiler Pro League.